Saisrivathsan Manikandan

+1 480-876-0016 | saisrivathsan2004@gmail.com | linkedin.com/in/smanika3 | github.com/smanika3 | smanika3.github.io

Professional Summary

Computer Science graduate student specializing in multimodal AI, LLM-based multi-agent systems, and deep learning for computer vision. Experienced in building AI pipelines with PyTorch, FastAPI, and AWS (Bedrock, AgentCore, SageMaker), including distributed training, on-premises inference, and guardrailed LLM workflows with deterministic verification.

Education

Arizona State University Tempe, AZ
Master of Science in Computer Science *Expected May 2027*
- **Relevant Coursework**: Cloud Computing, Data-Intensive Systems for Machine Learning, Data Mining

Arizona State University Tempe, AZ
Bachelor of Science in Computer Science, GPA: 3.59/4.00 *May 2026*
- **Relevant Coursework**: Data Structures & Algorithms, Image Processing & Analysis (Deep Learning), Machine Learning

Technical Skills

Languages: Python, C, C++, Java, SQL
AI/ML: PyTorch, Swin Transformer, Hugging Face, Computer Vision, CNNs, Scikit-learn, Generative AI, LLMs, Multi-Agent Systems, Strands Agents, Distributed Data Parallel (DDP), Pandas, NumPy
Backend & Cloud: FastAPI, Uvicorn, Asyncio, REST APIs, AWS (Bedrock, AgentCore, SageMaker, S3, Aurora PostgreSQL), CloudFormation, Linux
DevOps & Tools: Docker, Git/GitHub, GitHub Actions, Weights & Biases (W&B), TensorBoard, Pillow

Projects

Medical Imaging AI System with Swin Transformers | GitHub | *Python, PyTorch, Timm, TorchVision*
- Adapted published Swin Transformer methods to chest X-ray tasks: teacher-student learning with uncertainty-aware pseudo-labeling on CheXpert (14 pathologies, **0.9004 mean AUC**), TB lesion localization with FPN + Faster R-CNN (**mAP@0.5: 0.66**), and anatomical segmentation with Swin-UNet V2 (**Dice: 0.96**).
- Built a fault-tolerant multi-GPU PyTorch DDP training pipeline with mixed precision and automatic checkpoint resumption.

SlideSight | GitHub | Demo | *Python, FastAPI, Qwen3-VL, Gemma4, Pillow* **1st Place & Fan Favorite**
- Built an on-premise multimodal accessibility pipeline for the ASU AIR Spark Challenge using Qwen3-VL-32B-Instruct, FastAPI, and `python-pptx`, processing **380 images across 505 slides** from 10 university course decks at approximately **1.4 seconds per image** while keeping data on university hardware.
- Engineered a confidence-gated human-in-the-loop workflow combining model confidence, slide-text overlap checks, spatial occupancy, and Pillow color-histogram analysis; automatically generated **228 alt-text descriptions**, silenced **122 decorative graphics**, and escalated **30 ambiguous images** for review.
- Implemented a deterministic **WCAG 2.1 AA** audit engine for missing titles, typography, table headers, hyperlinks, and reading order, while using bounded asynchronous concurrency and a thread-safe FastAPI job registry to flag **531 accessibility violations**.

FundSentinel | GitHub | *Python, Bedrock AgentCore, Strands Agents, Aurora PostgreSQL, S3* **2nd Place**
- Built a 14-agent system on Strands Agents and Bedrock AgentCore (Claude Sonnet/Opus) for ETF and mutual-fund vetting (TIAA & ASU AI Investment Spark Challenge, team of 4): an LLM Supervisor routes Compliance, Analyst, Finance, and Suitability agents, and a code-plus-LLM evidence checker rejects any verdict whose cited numbers do not trace to source rows.
- Designed deterministic guardrails around every LLM step: schema mappings validated against data with up to 3 retries, repairs gated at 90% confidence and blocked if they flip a policy fail to a pass (**142 unsafe repairs** blocked), capped send-backs, and human escalation on unresolved cases.
- Implemented incremental re-review (reopens only affected agents: **6 of 32** reviews redone) and an LLM red-team harness (**11/13** trick funds caught, **0** false rejections); added CSV encoding/delimiter auto-detection and validated **15/15** remapped columns, **13/13** planted defects, and **23,783 funds** in under **5 minutes**.

Certifications & Training

AWS Certified AI Practitioner | Credential **MLOps Masterclass (Workshop)** | Certificate